

Oscillations in a charged cylinder

X22 (2022) — English translation

You may need the integral:

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \operatorname{artanh} \frac{x}{\sqrt{a^2 + x^2}} + C,$$

where $\operatorname{artanh} y$ denotes the inverse hyperbolic tangent of y .

A1	A disk with radius R is charged with surface charge density σ_R . Determine the potential $\varphi(y)$ at a point on the axis at a distance y from the center of the disk. The potential is zero at infinity.	0.50
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A2	Two such disks of radius R charged with surface charge density $\sigma_R > 0$ are parallel to each other. The distance between the centers of the disks is $2L$, the centers are located on the axis of the disks. In the equilibrium position there is a charge q with mass m , which can only move along the axis of the disks. Determine the angular frequency ω_1 of vibrations of such a charge. What is the charge sign?	1.00
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A3	Now this charge can only move in a perpendicular direction. Express the angular frequency ω_2 of oscillations in this case in terms of ω_1 . What is the charge sign now?	1.00
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B1	The lateral surface of a cylinder with radius R and length L is charged with surface charge density σ_L . Determine the potential at a point on the axis at a distance z from the center of one of the bases of the cylinder. The potential is zero at infinity.	1.00
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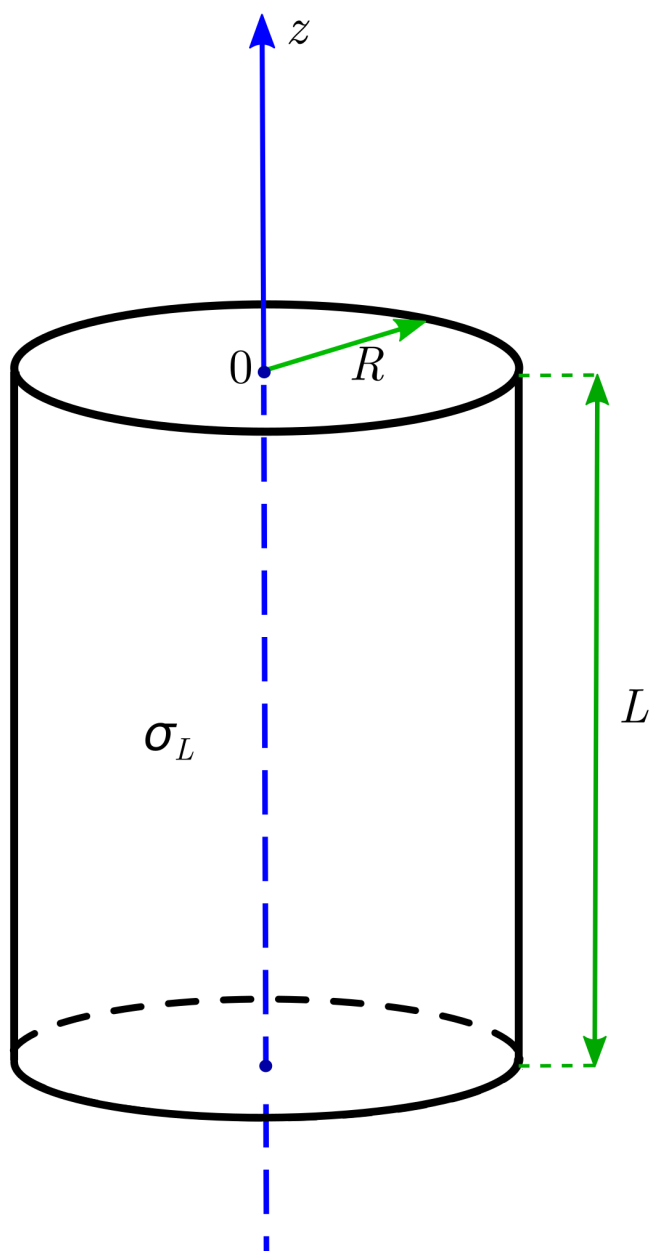


Figure 1: Figure 1.

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| B2 | Two such cylinders (with radius R and length L , surface charged with surface charge density $\sigma_L > 0$) are placed close together and have a common axis. In the equilibrium position there is a charge q with mass m , which can only move along the axis of the cylinders. Determine the angular frequency ω_3 of vibrations of such a charge. What is the charge sign? | 1.00 |
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B3 Now this charge can only move in a perpendicular direction. Express the angular frequency ω_4 of oscillations in this case in terms of ω_3 . What is the charge sign now? **0.50**

C1 A charged cylinder with radius R and height $L = 40R/9$ consists of a side surface and one base. Surface charge density of the side surface σ_L , base σ_R . If you place a point charge at the center of the opposite base, it will be in an equilibrium position. Define the relation σ_L/σ_R . **1.50**

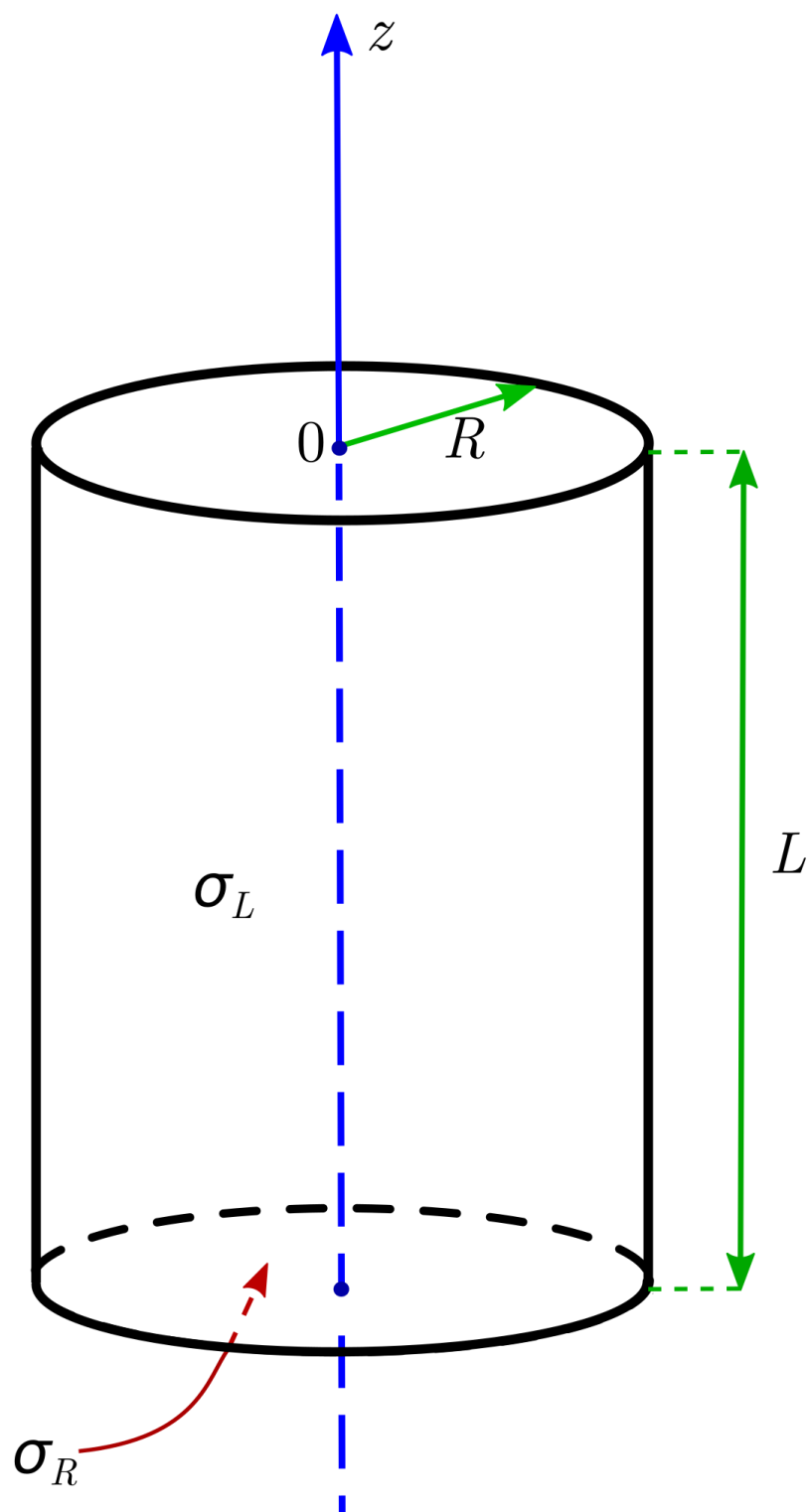


Figure 2: Figure 2.

C2 A charged cylinder with radius $R = 28b$ and height $L = 45b$ consists of a side surface and one base. Side surface charge $\sigma_L = -8\sigma_0$, base charge $\sigma_R = 25\sigma_0 > 0$. A particle with charge $q > 0$ is placed on the axis of this system. Estimate numerically the z coordinates (in b units) of the equilibrium positions if the particle can only move along the axis. Coordinate z is calculated as in the picture. Do this as accurately as possible, however, 1% accuracy is sufficient. Answers falling within 1% of correct will receive full marks. **2.50**

C3 Under the conditions of the previous paragraph, the particle was placed in the equilibrium position closest to the cylinder, its mass m . Determine the angular frequency ω of small vibrations of the particle. **1.00**

Source: <https://pho.rs/p/510>